

Skills Summary

Solving SAS and SSS Triangles with the Law of Cosines

Use this reference while completing your worksheet or when studying.

Skill 1 — SAS: the third side

$c^2 = a^2 + b^2 - 2ab \cos C$ — and the same shape for a or b . The side on the left is the one *opposite* the known angle.

Use it when you know two sides and the angle between them. If $C = 90^\circ$ the last term is zero and this is the Pythagorean theorem.

Example: $a = 7$, $b = 9$, included angle $C = 52^\circ$. Find c .

Step 1. Two sides with the angle between them, and no side has a known opposite angle — that rules out the law of sines and points straight at the law of cosines.

Step 2. $c^2 = 49 + 81 - 2(7)(9) \cos 52^\circ = 130 - 77.5733 = 52.4267 \Rightarrow c = \sqrt{52.4267} \approx 7.24$

Try it: $a = 6$, $b = 11$, included angle $C = 64^\circ$. Find c .

check: $c \approx 9.6$

Skill 2 — SSS: an angle from three sides

Rearranged: $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$. The side *opposite* the wanted angle is the one that gets subtracted.

Read the sign first: a negative cosine means the angle is obtuse, a positive one means acute. The largest angle always faces the longest side.

Example: $a = 6$, $b = 9$, $c = 11$. Find C .

Step 1. All three sides and no angles, so the law of sines has nothing to start from; rearranging the law of cosines for the cosine is the only route.

Step 2. $\cos C = \frac{36 + 81 - 121}{2(6)(9)} = \frac{-4}{108} = -0.0370$, which is negative, so expect an obtuse angle. $\Rightarrow C = \cos^{-1}(-0.0370) \approx 92.12^\circ$

Try it: $a = 7$, $b = 8$, $c = 12$. Find C .

check: $C \approx 109.9^\circ$

Skill 3 — SAS, then a remaining angle

Two steps. The law of cosines gives the third side; now that side and its opposite angle form a pair, so the law of sines finishes the triangle.

Safety rule: after finding the third side, aim the law of sines at the angle opposite the *shorter* of the remaining sides. That angle must be acute, so the inverse sine cannot mislead you.

Example: $a = 5$, $b = 8$, included angle $C = 66^\circ$. Find A .

Step 1. No side has a known opposite angle, so nothing can start with the law of sines — find the third side first, then a pair exists.

Step 2. $c^2 = 25 + 64 - 80 \cos 66^\circ = 56.4600$, so $c = 7.5141$; then $\sin A = \frac{5 \sin 66^\circ}{7.5141} = 0.6079$.
 $\Rightarrow A \approx 37.44^\circ$

Try it: $a = 4$, $b = 9$, included angle $C = 58^\circ$. Find A .

check: $A \approx 26.24^\circ$

Skill 4 — Distances and turns

Two routes leaving one point, with the angle between them, is SAS: the two route lengths are the sides and the angle at the shared point is included between them. The direct distance between the far ends is the third side.

Sanity check: an obtuse included angle makes the third side longer than either route; an acute one makes it shorter than their sum.

Example: Two tracks leave a gate, one $a = 250$ m long and one $b = 180$ m long, meeting at $C = 41^\circ$. How far apart are the far ends?

Step 1. The angle sits at the shared gate, between the two known lengths, so the direct distance is the side opposite it — SAS.

Step 2. $c^2 = 62500 + 32400 - 2(250)(180) \cos 41^\circ = 26976.1 \Rightarrow c \approx 164.24$ metres, shorter than either track, as an acute angle requires.

Try it: A triangular course has legs $a = 90$ m, $b = 140$ m and $c = 200$ m. Find the turn angle C opposite the longest leg.

check: $C \approx 119.22^\circ$